

**Course Code:**       EEE-1121  
**Course Title:**       Basic Electrical Engineering  
**Credit Hours:** 3  
**Contact Hours:** 3 lecture hours per week  
**Type:** Core, Engineering  
**Prerequisite:**       N/A  
**Co-requisite:**       EEE-1122 (Basic Electrical Engineering Lab)

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|--------------------|-----------------------------------------------|---------------------------------|----------|
| Course Assessments | <b>CIE:</b><br>Continuous Internal Evaluation | Attendance                      | 10 Marks |
|                    |                                               | Class test/ Assignment/ Quizzes | 10 Marks |
|                    |                                               | Mid-term                        | 30Marks  |
|                    | <b>SEE:</b> Semester End Examination          |                                 | 50 Marks |

This course introduces students to the basic laws, methods of analysis and theorems for direct current, DC and alternating current, AC circuit, such as, Ohms Law, Kirchhoff's, Current and Voltage Laws, Mesh and Nodal Analysis and Thevenin's and Norton's Theorems. Based on these, the students are expected to be able to solve for variables in any given DC and AC electric circuits. The students also exposed to the steady-state electrical circuit. Afterwards, the relevant concepts in three phase circuit analysis and passive RLC filter circuits are taught to the students. With the knowledge learned, the student would be able to apply the basic laws, theorem and methods of analysis for solving completely with confidence various problems in circuit analysis.

### Course Objective:

The objectives of this course are to

- familiarize with the fundamentals of electrical circuits
- learn to solve circuits by using different theorem
- learn the techniques to analyze the three-phase ac circuit

**Course Content:****Section-A (Mid-term: 30 Marks)**

**1. Circuit Elements and Basic Laws:** D.C voltage, current, resistance and power; dependent and independent sources, active and passive elements, networks definitions, conversion between Y and  $\Delta$  networks, Reduction of complicated networks, KVL & KCL.

**2. Network Analysis:** Elementary network topology, network constraints, and network equilibrium equations. Nodal and mesh network analysis, Supernode, Supermesh, Source Transformation.

**3. Network Theorems:** Superposition theorem, the reciprocity theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem.

**Section-B (Final Exam: 50 Marks)****Group-A (20 Marks)**

**4. Signal and Waveforms:** Signal waveforms: D.C., step, impulse, square pulse, sinusoidal, triangular, and exponential. General description of signals: time constant, rms value, duty cycle, crest factor, form factor.

**5. A.C Circuits:** Instantaneous, Average and R.M.S. values of current, voltage and power, Real, Reactive and Apparent power, Volt-ampere, Phase and Phase difference, Power Factor, Impedance, Determination of Average and R.M.S values of different sinusoidal and non-sinusoidal waves, power calculation of RL, RC and RLC circuits, Nodal and Mesh Analysis.

**Group-B (30 Marks)**

**6. Phasor Algebra:** Use of complex quantities in AC circuits, Phasor Diagram, Nodal and Mesh Analysis using Phasor, resonant circuits, Q value and band width, frequency response.

**7. Filters:** Different types of filters, Fundamental concepts of RL and RC filter, Properties of symmetrical networks filter fundamentals.

**8. 3- $\phi$  Circuits:** Introduction to Polyphase Circuits, three-phase three wire system, Y and  $\Delta$  connected system, Power calculation in balanced 3- $\phi$  systems.

**Text Book:**

1. C.K. Alexander and M.N.O. Sadiku, Fundamental of Electric Circuits, 5th Edition, McGraw-Hill 2009
2. I. Robert L. Boylestad, Introductory Circuit Analysis, 11th ed., Prentice Hall, 2006.

**Reference Books:**

1. J. Nilsson and S. Riedel, Electric Circuits 8th ed. Prentice-Hall, 2007
2. R. C. Dorf and J. A. Svoboda, Introduction to Electric Circuits, 7th ed, John Wiley, 2006
3. B. L. Theraja, A Textbook of Electrical Technology Volume-1, 23rd Edition, S. Chand, 2008  
Nannapaneni Narayana Rao, Fundamentals of electromagnetics for Electrical and Computer Engineering, Prentice Hall, 2009